

Flowing with Zen: Exploring Empowering the Intangible Cultural Heritage via Immersive Mixed Reality Spaces

Wenchen Guo^{1,2*} Guoyu Sun² Wenbo Zhao² Zhirui Chen² Menghan Shi³ Yu Zhang⁴ Chuqiao Wan⁵

¹Peking University, China; ²Communication University of China, China; ³Lancaster University, United Kingdom

⁴Tsinghua University, China; ⁵Imperial College London, United Kingdom

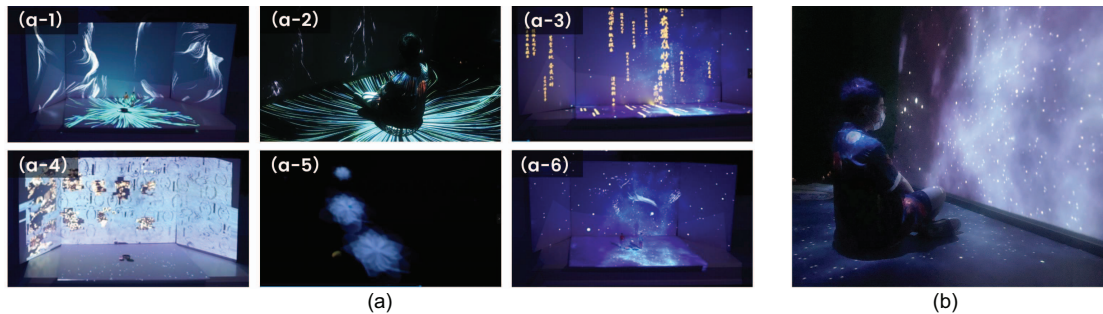


Figure 1: (a) The overview of *Flowing with Zen* we presented. (b) Offline exhibition.

ABSTRACT

Zen is a treasure of the world's intangible cultural heritage (ICH), but nowadays it is facing difficulties in dissemination. This paper presents an immersive experience space *Flowing with Zen* by integrating HCI technology and mixed reality (MR). Audiences can explore and interact with the four scenarios, as well as meditate, and experience Zen philosophy. The pilot study shows that the MR space not only evokes users' interest and participation but also deepens their empathy or reflections. This innovative way of combining MR, ICH, and experience enhances the accessibility of Zen, and it may open up a new mode of dissemination for ICH.

Keywords: Zen culture, intangible cultural heritage, human-computer interaction, immersive mixed reality space, user experience

Index Terms: Human-centered computing [Human computer interaction (HCI)]; Interaction paradigms—Mixed reality; Human centered computing [Interaction design]; Interaction design process and methods—Scenario-based design

1 INTRODUCTION

Zen culture holds a significant positions in Chinese society, exerting profound influences on Chinese culture as a whole and on each individual. As a spiritual belief and practice, Zen emphasizes mindfulness and contemplation, encouraging individuals to focus on inner awareness and experiences, and by engaging in meditation and Zen practice, they free themselves from trivialities, achieving a balance between body and mind. This holds great significance for the Chinese people who are currently immersed in a fast-paced and high-pressure era.

However, Zen culture faces challenges in terms of

dissemination and inheritance. On the one hand, people's ways of obtaining information and experiencing culture have changed, particularly in the context of new media. On the other hand, the concepts of them often possess certain levels of abstraction and complexity, making them difficult for the general public to comprehend, thus reducing the accessibility of the culture. In recent years, HCI technologies such as immersive media (VR/AR/MR) have seen a boom. Many studies have shown that the intervention of these technologies has revitalized many tangible cultural heritages (TCH) [1, 2], but there are still fewer explorations on empowering intangible cultural heritages (ICH). *How to design and develop appropriate immersive environments for ICH (e.g. Zen)? Can immersive environments enhance the motivation of the younger generation to participate and provide them with the desired user experience and learning outcomes?* These questions remain to be addressed.

In this short paper, our core work is to explore this possibility. Specifically, based on the study of relevant Zen cultural scriptures and core concepts, we have designed and developed a new media art space called *Flowing with Zen* (see Fig. 1). It presents four scenarios in a temporal sequence, providing an interactive and exploratory space for the audience to experience philosophy and meditation for relaxation of body and mind. In addition, we have conducted a pilot study. Preliminary results indicate that this space not only evokes the audience's sense of participation, immersion, and imagination but also deepens their resonance and contemplation.

2 IMPLEMENTATION OF FLOWING WITH ZEN

2.1 Scenarios and Interaction Design

The *Flowing with Zen* presents 4 themed scenarios: “Wandering” “Emergence” “Seeking” and “Perfection”, following the sequence of time. Audiences will experience the immersive transition between these scenarios in the space, which also serves as a metaphor for the journey of life and the trajectory of existence. The aim is to make the audience perceive the flow of time and contemplate philosophical questions related to the cycles of life.

* Corresponding author: beihuanfanchen@sina.com

Scenario 1: Wandering. “Life flows like a waterfall, never ceasing, never stopping.” The concept of *Wandering* refers not only to the thread-like symbols in space, but also to the trajectories of all living beings in the world. The visual effects are generated in real-time by computer programs. Interactive Description: As the audience moves within the space or as the number of audiences changes, the quantity, generation speed, and randomness of the trajectory lines will also vary accordingly. (Please see Fig. 1-a-1) Additionally, the floor projection will be triggered when the audience sits on a cushion in the center of the space and meditates (Please see Fig. 1-a-2).

Scenario 2: Emergence. “Never shall there be the names of suffering, hardships, evil realms, or demonic afflictions.” In this scenario, we extract, transform, and process profound and inspiring scriptures into perceptible stereoscopic visual scenes, presented in the form of a continuous rain of words (Please see Fig. 1-a-3 for the overall effect). Interactive Description: When the audience approaches, reads, and touches the scriptures on the wall, the scriptures gradually fade away.

Scenario 3: Seeking. “One Flower, One World, One Leaf, One Bodhi.” *Seeking* is the core scenario. In this scenario, the audience can feel and explore freely. Interactive Description: When the audience comes to the stone wall, reaches out, and touches, the stone wall automatically peeled off, and the back of the mural (the picture of the Western Pure Land) gradually appeared (Please see Fig. 1-a-4); when the audience walks, lotus flowers accompany each other, creating the Zen beauty of “step by step, a lotus flower grows” (Please see Fig. 1-a-5).

Scenario 4: Perfection. “The endless cycle of birth and death is eternal at this moment.” The *Perfection* is the final scenario. After the dissipation of the imagery of the Pure Land in the previous scenario, Scenario 4 guides the audience’s vision back to the universe and void. The visual symbol of the “Seal of Wishes” appears, expressing the meaning of “fulfilling the wishes of all beings and enabling the realization of their desired aspirations” (Please see Fig. 1-a-6 for the overall effect). Interactive Description: When the audience comes to the center of the space, the “Seal of Wishes” moves to the feet of the audience.

2.2 Technical Framework

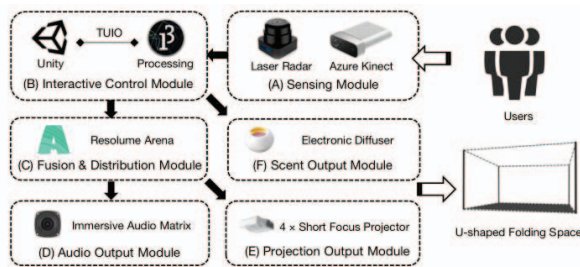


Figure 2: Technical framework diagram for *Flowing with Zen*.

Fig. 2 shows the technical framework and workflow of *Flowing with Zen*. The system mainly consists of Sensing Module, Interaction Control Module, Fusion and Distribution Module, three Output Module (Audio, Projection, and Scent), and a PC (high-performance graphics workstation).

After the user(s) enters the interactive area, (A) Sensing Module identifies, collects the number, position, and action data of the user(s) through radar and Kinect (body camera), and transmits it to (B) Interactive Content Control Module. (B) generates interactive effects according to pre-defined rules and

transmits the content through the Syphon / Spout protocol to (C) Fusion & Distribution Module and (F) Scent Output Module (driven by Arduino). (C) fuses, divides, corrects, and distributes the scenes according to the size and shape of the actual exhibition space, and transmits them to (D) Audio Output Module and (E) Projection Output Module. (D), (E), and (F) output digital content and scent in the exhibition space (U-shaped folding space).

3 PILOT STUDY

During the offline exhibition (see Fig. 1b), we conducted a 3-day pilot study, involving a total of 66 audiences (Aged 19–72, Mean = 30.4, SD = 11.1, Females = 40). We used a mixed-methods (i.e., questionnaire measurements and brief interviews) to measure audiences’ satisfaction, personalized experience, and subjective feelings [2, 3].

Preliminary survey results show that *Flowing with Zen* is recognized by audiences of different cultural backgrounds and ages in terms of satisfaction ($M = 3.88$) and personalized experience ($M = 4.11$). There were many respondents who said, “...my heart was cleansed” “...it made me not help but slow down and reflect on my inner world”. It is worth mentioning that some of the young audience also took photos or videos of the experience and shared them with their feelings on social media, promoting the secondary dissemination of Buddhism and Zen culture. However, there is still room for optimizing the interactive design ($M = 3.46$), such as providing interactive cues during the process, controlling the number of audience present at the same time to provide a better individual experience, etc.

4 CONCLUSION & FUTURE WORK

This study is dedicated to creating an immersive experiential space that is connected to Zen culture through HCI and new media art. It provides the audience with a deeper opportunity to experience and connect with it. The pilot study also showed that our success in integrating Zen culture with the aesthetics and context of contemporary society was recognized by the audience. We believe that this innovative way of combining technology, experience and intangible cultural heritage enhances the accessibility of traditional culture and has the potential to be a highly promising mode of dissemination of intangible heritage.

In the future, we will continue to refine our work and explore more possibilities for the integration and development of intangible cultural heritage and technology.

ACKNOWLEDGEMENTS

The work is supported by the Fundamental Research Funds for the Central Universities (CUC23GZ006).

REFERENCES

- [1] Y. Wu, Y. Huang, W. Guo, X. Li, Z. Chen, Z. Feng, Y. Zhu, and G. Sun. Multi-sensory Interactive Experience Design of Museum Musical Instrument Collections. In *HCI International 2023 - Late Breaking Papers*. Cham: Springer Nature Switzerland, 2023, pp. 420–437.
- [2] W. Guo, Y. Huang, Z. Chen, Z. Zhang, G. Sun, Q. Zeng, and X. Li. The “rebirth” of traditional musical instrument: An interactive installation based on augmented reality and somatosensory technology to empower the exhibition of chimes. *Computer Animation and Virtual Worlds*, vol. 34, no. 3, pp. e2171, 2023.
- [3] S. Jin, M. Fan, and A. Kadir. Immersive Spring Morning in the Han Palace: Learning Traditional Chinese Art Via Virtual Reality and Multi-Touch Tabletop. *International Journal of Human-Computer Interaction*, 2021, pp. 1–14.